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SPECIFICATION — Model Playground (PyQt5 + uv)

- **Status:** v0.1 — draft for implementation
- **Language:** Python 3.12 | GUI: PyQt5 | HTTP: httpx | Schema: jsonschema
- **Curriculum source:** `curriculum/week1/chapter1.md` (§12 Model Playground, §13 Measuring Tokens, §14 Comparing Outputs, §15 Enforcing Structured JSON).
- **Scope of this document:** the *authoritative specification* of an AI-native system. It is written to Level 2–3 (structured, mostly executable): behavior, interfaces, invariants, edge cases, and failure semantics are made explicit so an agent (or engineer) can derive implementation and verification with minimal inference.
- **Principle:** requirements express *intent*; this specification *operationalizes* intent into observable behavior plus the conditions under which we know it is correct.

0. Intent and purpose

Build a small desktop application that treats **different LLMs as interchangeable computational components** sitting behind one common interface, so that an engineer can run the same prompt through several models **side by side** and inspect, per model, not just the text but the *inference-substrate* properties that actually determine application quality:

- **latency** — decomposed into time-to-first-token (TTFT) and total latency;
- **throughput** — generation rate (tokens/second, TPS);
- **economics** — token usage and derived *cost*. Locally there is no vendor bill, so cost is a configurable per-1k-token field for bookkeeping that defaults to 0; the machinery is identical to the hosted case (chapter §13).
- **reliability** — does the output parse, and does it satisfy a schema?

This directly instantiates the chapter’s core thesis: *AI Application = Probabilistic Components + Deterministic Systems*. The model supplies the probabilistic computation (§17); everything here is the **deterministic boundary** — the “harness layer” (§1) and the “reliability boundary” (§15) — that the model is never trusted to provide on its own.

Deployment decision (chapter §3, APIs vs. Local Models): inference is *local*, performed by the **Ollama** runtime (default `http://localhost:11434`) against models already pulled on the machine. The app talks to Ollama’s HTTP API (a small `localhost` HTTP call, not a hosted cloud API) and the model runs on the host’s CPU/GPU/NPU. This is the “local inference” architecture of chapter §3: the app gains control over latency, privacy, availability, version, and configuration, and pays the corresponding engineering burden (owning runtime availability, model download, memory/thermal constraints — see [E-13/E-14/E-15](#)).

Primary product surface: not “chat with a model,” but a **side-by-side evaluation panel** in which, for one prompt, the user sees $M_1(x), M_2(x), \dots$ with each model’s metrics and — optionally — its validated structured output.

Non-goals (explicit, to constrain the solution space):

- No conversation history / multi-turn memory beyond the current prompt (§6 says history is a *context* input, not a feature to build here).
- No tool calling / function-call execution loop (§7 is conceptual; out of scope for this minimal substrate). *See open question Q-04.*
- No multimodal inputs (§8 is conceptual; text-only here).
- No persistence of runs, no cloud/hosted provider, no auth. Ollama is a `localhost` daemon with no key ([E-13](#)). Model **download/pull** is out of scope — the app lists and runs *already-pulled* local models (§3 local-inference model, [E-14](#)).
- No alternative “quality” LLM-as-judge scoring: comparison here is *observational* (metrics + human/JSON inspection), not an automated judge.
- **The app must not require Ollama to import, build, or run its test suite:** a deterministic **MockModel test double** provides every capability for offline/CI use (§9). Ollama is the *real* backend; **MockModel** is what the test suite drives.

1. Actors and goals

Actor	Goals
User (human, single process)	Enter one prompt; pick a set of models + generation parameters; run them in parallel or sequentially; compare outputs, metrics, and costs side by side; optionally obtain and inspect a validated structured output.

Actor	Goals
Model (Model implementation: <code>OllamaModel</code> for the real run, <code>MockModel</code> as a test double)	Given messages + params, produce a (streaming) text completion with usage stats, or fail — never touch the GUI.
Ollama daemon (<i>external</i>)	Local inference runtime on <code>localhost:11434</code> (<code>/api/chat</code> , <code>/api/tags</code>). Owns the model weights and the CPU/GPU/NPU. Not part of this project; E-13/E-14 handle its absence.
Registry (<code>ModelRegistry</code>)	Map a model identifier to a configured model + its pricing, so the rest of the system never branches on provider. For Ollama, populated from <code>GET /api/tags</code> .
Metrics layer (<code>metrics.py</code>)	Turn a run’s raw timings + usage into TTFT / latency / TPS / cost. Pure, headless, testable.
Structured layer (<code>structured.py</code>)	Turn a raw text completion into a <i>validated</i> typed object via <code>parse</code> → <code>validate</code> → <code>accept/reject</code> , with a <code>retry/fallback</code> policy. Pure, headless, testable.
Worker (<code>RunWorker(QThread)</code>)	Drive one model’s generation off the UI thread; emit stream chunks and a final result; crash cleanly.
UI (<code>MainWindow</code>)	Present controls + a grid of per-model result panels; never block on inference.

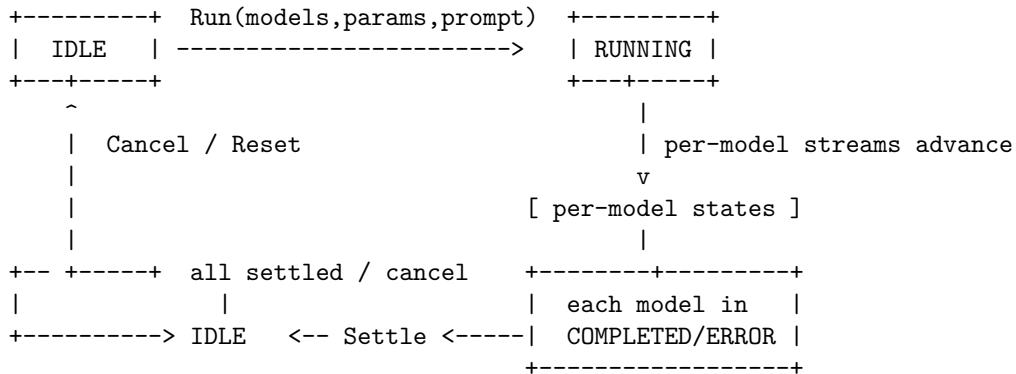
2. Requirements (intent, high level)

ID	Statement
R-01	The application shall expose a common model interface (<code>Model</code>) such that a model can be substituted for another without changing any other layer of the system.
R-02	The user shall specify, per run: a list of one or more models (chosen from the locally-pulled Ollama models, §R-16, or from the built-in <code>MockModel</code> registry when Ollama is unavailable), a generation-parameter set (temperature, <code>top_p</code> , <code>max_tokens</code>), and the prompt (as a message list).
R-03	For each selected model the app shall display the generated text (fully, or streaming token-by-token).
R-04	For each model run the app shall display metrics : <code>tput</code> latency (ms), <code>ttft_ms</code> , <code>tps</code> , and <code>status</code> .
R-05	For each model run the app shall record token usage : <code>prompt_tokens</code> , <code>completion_tokens</code> , <code>total_tokens</code> . For Ollama these come from the response envelope (<code>prompt_eval_count</code> , <code>eval_count</code>); for <code>MockModel</code> from its deterministic tokenizer.
R-06	For each model run the app shall display a cost computed from usage and that model’s registered per-1k-token prices. For local Ollama prices default to 0 (no vendor bill); the field is configurable so the same machinery models a nominal compute cost (chapter §13).
R-07	The app shall run the selected models side by side , i.e. the user can compare $M_1(x), \dots, M_k(x)$ from one prompt in a single view (§14).
R-08	Streaming mode shall show the response as it arrives and must report $TTFT \leq T_{complete}$ and distinguish them (§12.2, §11).
R-09	When structured output is requested, the app shall emit a validated typed object (not raw prose): { <code>answer</code> , <code>confidence</code> , <code>reasoning_required</code> } (or the configured schema).
R-10	Structured output shall enforce the pipeline raw → parse → validate → accept/reject ; never assume “valid-looking ≥ <i>valid</i> ” (§6, §15).

ID	Statement
R-11	On a parse/validation failure the app shall apply a retry-then-fallback policy and surface the failure state, never silently fabricate a valid object (§15).
R-12	The GUI must remain responsive while one or more models generate (§3 streaming).
R-13	All model I/O must occur off the Qt main (event-loop) thread.
R-14	The project shall be reproducible via uv on Python 3.12, and runnable/testable fully offline (no Ollama) via MockModel; the real GUI uses Ollama when reachable and degrades gracefully when it is not (E-13).
R-15	A fixed seed + temperature=0 shall make a run deterministic : MockModel is <i>bitwise</i> reproducible; OllamaModel is reproducible <i>best-effort</i> (Ollama honors <code>options.seed</code> , but floating-point/kernel differences across runs may perturb later tokens — token <i>counts and metrics</i> are asserted, bitwise text is not).
R-16	On launch the app shall discover locally-pulled Ollama models via GET <code>/api/tags</code> and populate the model checklist from that list; if Ollama is unavailable it shall fall back to the built-in MockModel registry and state so in the UI (E-13).
R-17	A model that is not pulled locally shall not be runnable: selecting it yields a per-panel ERROR (model not found / pull required) rather than a silent failure or a hard crash (E-14).

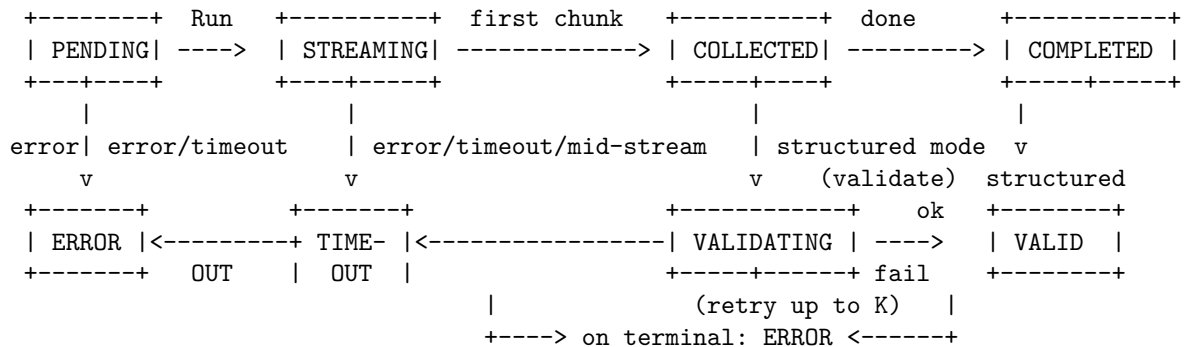
3. Behavior and state model

3.1 System state machine (RunState, per comparison run)



Overall run state is an **aggregate** of the per-model states. A run is settled when *every* model has reached COMPLETED or a terminal error. A single model failure never aborts the others.

3.2 Per-model run state machine (ModelRunState)



State	Meaning	Terminal?
PENDING	Scheduled; request not yet issued.	no
STREAMING	Chunks arriving; <code>text</code> is a partial accumulation.	no
COLLECTED	Full text + usage available (streaming done, or non-streaming returned).	no
VALIDATING	Structured mode: parse + schema-validate the collected text; may retry.	no
COMPLETED	Plain-text mode terminal success (no structured request).	yes
VALID	Structured mode: a validated typed object exists.	yes
ERROR	Terminal fault: request error, timeout, mid-stream failure, or all retries exhausted (§8).	yes
TIMED_OUT	A dedicated terminal when the total elapsed exceeds <code>timeout_s</code> .	yes
CANCELLED	User cancelled the run while this model was active.	yes

Transition rules: - Exactly **one** attempt sequence per model per run; within it, structured mode may perform up to `max_retries` parse attempts (§8). Each retry re-issues generation with an *error-informed* prompt (original prompt + a directive describing the last parse/validation failure). - A model reaching **ERROR**/**TIMED_OUT**/**CANCELLED** is terminal for *that* panel but leaves all sibling panels running. - The overall run reaches **IDLE** (settled) only when **every** panel is terminal. - **Stop/Cancel** from **IDLE** while a run is active tears every model's worker down (no live worker survives; see [I-010](#)).

3.3 Threading model

- **UI thread (main):** widget event loop + all panel updates only.
- **Worker thread(s):** one `RunWorker(QThread)` **per model**, each owning its `Model` and its `generate/stream` call. Never touch widgets.
- Communication is Qt signals only (`QThread` + `pyqtSignal`), plus a per-worker thread-safe cancel flag guarded by a lock. No shared mutable buffers cross threads except via queued (auto-connect) signals.
- **Concurrency policy:** by default all selected models stream **concurrently** (one worker each). A sequential mode (one-at-a-time) is allowed via UI flag to keep `concurrency = 1`; default is **concurrent**.

3.4 Streaming vs. non-streaming

The common interface offers **both**; the UI picks one:

- **Streaming:** the model yields `StreamChunks`; `ttft` is measured to the **first non-done chunk** that carries a non-empty delta. Ollama emits **NDJSON** — one JSON object per line `{"message": {"role","content"}, "done": bool, ...}`; `OllamaModel` splits the response on newlines and parses each line; the **final** line has `done: true` and carries `prompt_eval_count / eval_count` (the [C-01 Usage](#)). `MockModel` yields equivalent `StreamChunks` with no HTTP.
- **Non-streaming:** `OllamaModel` sets `stream=false` and parses the single JSON object; the interface returns one `ModelResponse`; `ttft = total_latency` (a special-case defined explicitly so the metric is always total-ordered, [E-05](#)).
- **Determinism:** generation passes `options.seed` and `temperature`; with `temperature=0` + fixed `seed` Ollama output is reproducible ([R-15](#), best-effort).

4. Interfaces / contracts

C-01 Core types

```
class Role: SYSTEM = "system"; USER = "user"; ASSISTANT = "assistant"

@dataclass
class Message:
    role: str          # one of Role
    content: str

@dataclass
class GenerationParams:    # part of "generation parameters" the model needs
    temperature: float = 0.0 # in [0, 2]; 0 => deterministic-ish
    top_p: float = 1.0 # in (0, 1]
    max_tokens: int = 512 # > 0
    seed: int | None = None # fixed => reproducible when the model supports it (R-15)

@dataclass
class Usage:
    prompt_tokens: int # >= 0
    completion_tokens: int # >= 0
    @property def total_tokens(self) -> int ... # = prompt + completion (I-001)

@dataclass
class StreamChunk:
    delta: str          # text delta; may be "" on a keepalive
    finished: bool      # True on the final chunk
    usage: Usage | None # set (or updated) on the final chunk; None on mid-stream

@dataclass
class ModelResponse:    # the non-streaming return, and the *collected* form
    text: str
    usage: Usage
    model_id: str
```

C-02 Model — the minimal interchangeable interface (R-01)

```
class Model(ABC):
    @property
    def model_id(self) -> str: ... # stable id, e.g. "/ollama/llama3.2", "mock/fast"

    @abstractmethod
    def generate(self, messages: list[Message], **params) -> ModelResponse:
        """Non-streaming. Precondition: messages non-empty.
        Postcondition: return a ModelResponse whose usage.prompt_tokens >= 0.
        May raise on transport/timeout errors (caught by the worker -> ERROR)."""

    @abstractmethod
    def stream(self, messages: list[Message], **params) -> Iterator[StreamChunk]:
        """Streaming. Yields >=1 chunk; the LAST chunk has finished=True and
        carries the final usage (I-008)."""
```

****params** accepts `GenerationParams` fields. **Invariance (I-002)**: no layer other than a concrete `Model` implementation (`OllamaModel`, `MockModel`) may hard-code a provider endpoint, model name, or provider-specific request shape. All other modules see only `Model` + `Message` + `GenerationParams`.

C-03 ModelRegistry

```
@dataclass
class ModelSpec:
    model: Model
    label: str | None = None           # display name; defaults to model.model_id
    price_input_usd_per_1k: float = 0.0
    price_output_usd_per_1k: float = 0.0

class ModelRegistry:
    def __init__(self) -> None ...
    def register(self, spec: ModelSpec) -> None
    def get(self, model_id: str) -> ModelSpec      # KeyError if unknown
    def available(self) -> list[ModelSpec]
    def cost_usd(self, model_id: str, usage: Usage) -> float
```

Invariance (I-003): pricing is bound to the model in the registry, not hard-coded in the UI or the worker. For local Ollama, `price_output_usd_per_1k` and `price_input_usd_per_1k` default to 0.0 (no vendor bill, R-06); the fields remain so a nominal compute cost can be booked in.

C-03b OllamaClient (thin transport for OllamaModel, model.py)

A tiny `httpx`-based client over the Ollama HTTP API. It is the *only* place that knows URLs/JSON shapes (I-002). No import of a vendor SDK; `httpx` is a generic HTTP client.

```
OLLAMA_BASE = "http://localhost:11434"    # overridable via env OLLAMA_HOST
```

```
class OllamaClient:
    def __init__(self, base: str = OLLAMA_BASE, timeout_s: float = 30.0) -> None ...

    def list_models(self) -> list[str]:
        """GET /api/tags -> [name, ...] of *locally-pulled* models (R-16, E-13).
        Raises ConnectionError-like on unreachable daemon; the caller treats that as
        "Ollama unavailable" and falls back to MockModel (E-13)."""

    def chat(self, model: str, messages: list[Message],
             params: GenerationParams, stream: bool) -> "Iterator[StreamChunk] | ModelResponse":
        """POST /api/chat. Maps GenerationParams -> Ollama `options`
        (temperature, top_p, num_predict=max_tokens, seed). Stream: NDJSON, last
        line carries prompt_eval_count/eval_count -> Usage (R-05, I-008).
        Raises on HTTP/network error or malformed NDJSON (E-02/E-07/E-15)."""
```

Mapping (GenerationParams -> Ollama options): `temperature`→`temperature`, `top_p`→`top_p`, `max_tokens`→`num_predict`, `seed`→`seed` (omit when `None`). **Endpoint for unknown model**: Ollama returns HTTP 404/error for a model not pulled locally; `OllamaModel` surfaces that as `KeyError`/panel ERROR (R-17, E-14).

C-04 Metrics (metrics.py, pure — testable headless)

```
@dataclass
class RunMetrics:
    model_id: str
```

```

status: str
ttft_ms: float          # ms to first non-empty delta (or total latency if non-streaming)
total_latency_ms: float
tps: float              # completion_tokens / generation_seconds
usage: Usage
cost_usd: float
retries: int = 0         # structured-mode parse retries that occurred
error: str | None = None

def compute_metrics(
    model_id: str,
    *,
    t_request: float, t_first_token: float | None, t_complete: float,
    usage: Usage, price_input: float, price_output: float,
    retries: int = 0, status: str = "COMPLETED", error: str | None = None,
) -> RunMetrics

```

Definitions (enforced by I-004, I-005, I-006):

$$T_{\text{ttft}} = t_{\text{first token}} - t_{\text{request}} \quad T_{\text{complete}} = t_{\text{complete}} - t_{\text{request}}$$

$$\text{TPS} = \begin{cases} \frac{\text{completion_tokens}}{T_{\text{complete}} - T_{\text{ttft}}} & \text{if } T_{\text{complete}} > T_{\text{ttft}} \\ 0.0 & \text{otherwise (guard, I-005)} \end{cases}$$

$$\text{cost_usd} = \frac{\text{prompt_tokens}}{1000} P_{\text{in}} + \frac{\text{completion_tokens}}{1000} P_{\text{out}} \quad (\text{C-13 unit economics})$$

- **Cost per successful task** (§13) is a *derived* aggregate the UI shows: total cost across a run divided by the count of models that reached a terminal success state (COMPLETED or VALID). If that count is 0, the denominator is 1 and the value is the raw total (never division by zero; I-007).

C-05 Structured output (structured.py, pure — testable headless)

Target schema (the default “answer” schema of §15):

```

{
  "type": "object",
  "additionalProperties": false,
  "required": ["answer", "confidence", "reasoning_required"],
  "properties": {
    "answer": {"type": "string", "minLength": 1},
    "confidence": {"type": "number", "minimum": 0, "maximum": 1},
    "reasoning_required": {"type": "boolean"}
  }
}

@dataclass
class ValidationResult:
    ok: bool
    data: dict | None      # populated iff ok
    errors: list[str]      # human-readable reasons; empty iff ok
    raw: str               # the exact text that was parsed

def parse_json(text: str) -> tuple[dict | None, list[str]]:

```



```

"""Strip a single ```json ...``` fence if present, then json.loads.
Returns (None, [reason]) on any failure - never raises."""

```

```

def validate(data, schema) -> ValidationResult:
    """jsonschema.validate; collect every error message. Never raises."""

```

Pipeline (structured.py):

```

raw text --> strip optional code fence --> json.loads --> jsonschema.validate --> dict
\_____ any step fails => attempt retry (up to max_retries),
           else terminal ERROR with the first failure reason (E-03)

```

C-06 RunWorker(QThread)

signals:

```

    token(model_id: str, delta: str)      # per streamed chunk (UI appends)
    metrics_ready(model_id: str, RunMetrics) # emitted once, on settle or terminal
    crashed(model_id: str, message: str)   # uncaught worker fault (E-07)

```

public methods (thread-safe to call from main):

```

    start_run(...) -> None      # kick off the model's generate/stream
    cancel()        -> None     # idempotent; safe from any state

```

Each RunWorker is 1:1 with one model panel. A comparison run owns a QThreadList-style container managing N workers concurrently.

C-07 Data structures carried to the UI

ModelPanelView:

```

    model_id, label, status, text (accumulated),
    metrics: RunMetrics | None,
    structured: ValidationResult | None,
    streaming: bool, done: bool

```

5. UI specification

5.1 Top-level layout (MainWindow)

+-----+ Title: Model Playground - an inference substrate -----+		
LEFT COLUMN (controls)	RIGHT COLUMN: grid of per-model panels	
Prompt (QPlainTextEdit)	+-----+	
system prompt (opt.)	[Mock Fast]	RUNNING
Temperature / top_p /	text: "..." (streaming)	
max_tokens / seed	TTFT __ ms	Lat __ ms TPS _
Models: checklist of	in __/out __ cost \$0.000	
registered models (>=1)	[structured: OK/FAIL badge]	
[stream] checkbox	+-----+	
[structured] checkbox	+-----+	
[sequential] checkbox	[Mock Slow]	COMPLETED
[Run][Cancel]	text: "..."	
state label	TTFT __	Lat __ TPS _ \$__
cost-per-task summary	+-----+	
+-----+		

C-08 Control widgets and validation

Widget	Type	Constraint	Invalid behavior (§8)
Prompt	QPlainTextEdit	non-empty	disable Run; message Prompt must not be empty
Temperature	QDoubleSpinBox	[0, 2]	clamp
top_p	QDoubleSpinBox	(0, 1]	clamp; message if out of range
max_tokens	QSpinBox	1 ... 100,000	message if < 1
seed	QLineEdit(int, optional)	integer or blank	non-integer -> message + disable Run; blank => None
Models	QCheckBox per registered model	≥ 1 selected	disable Run; message Select at least one model
Run / Cancel	QPushButton	enabled per state table (§3.1)	–
stream / structured / sequential	QCheckBox	–	–

5.2 Panels

- Each panel shows: label, status pill, growing text, the four C-04 metrics, usage in/out, cost, and — when structured mode is on — a pass/fail badge plus the parsed JSON or the collected validation errors.
- Streaming: text appends as `token` signals arrive; metrics update once on settle.

6. Invariants (must hold in every valid implementation)

ID	Invariant	Verified by
I-001	<code>Usage.total_tokens == prompt_tokens + completion_tokens</code> ; both ≥ 0.	T-01
I-002	No module outside a concrete <code>Model</code> implementation imports a provider-specific SDK or names a concrete model. (Architecture; checked by T-02 import scan.)	T-02
I-003	Pricing lives only in <code>ModelRegistry</code> ; <code>cost_usd</code> is derived, never hard-coded in UI/worker.	T-03
I-004	<code>ttft_ms ≤ total_latency_ms</code> for every finished run (incl. the non-streaming special case <code>ttft = total</code>).	T-04
I-005	<code>tps ≥ 0</code> ; the zero-generation-duration case yields <code>tps == 0.0</code> (never <code>inf/nan</code>).	T-04
I-006	<code>cost_usd</code> equals the C-04 formula exactly (within float epsilon) for the supplied prices.	T-05
I-007	<code>cost_per_success_task</code> is never a division by zero; all-failed run uses denominator 1.	T-06
I-008	Every <code>stream</code> yields a final chunk with <code>finished=True</code> ; that chunk's <code>usage.completion_tokens</code> equals the streamed token count and equals the non-streaming <code>generate</code> usage for the same input+seed (mock, R-15).	T-07

ID	Invariant	Verified by
I-009	No valid-looking output is accepted: a panel reaches VALID only via <code>validate(...).ok == True</code> . A syntactically valid but out-of-range/missing-field object must NOT validate.	T-08
I-010	At most one live worker per model panel; <code>cancel()</code> leaves zero live workers.	T-09
I-011	All heavy inference is off the UI thread (worker thread id != main thread id).	T-11
I-012	Fixed <code>seed</code> on the mock model yields a bitwise-identical <code>text</code> , <code>usage</code> , and metrics for the same prompt + params.	T-07

7. Constraints (precise and measurable)

ID	Constraint	Measurement
K-01	While RUNNING with all models streaming, a posted task on the event loop is serviced within $p_{95} < 50$ ms.	T-11 (offscreen)
K-02	A mid-stream failure or timeout leaves the panel in a well-defined terminal ERROR/TIMED_OUT ; partial text is preserved for display.	E-02 , T-10
K-03	Default <code>max_retries</code> = 2; default per-run <code>timeout_s</code> = 30.	T-08
K-04	No provider SDK or key is required to import, run, or test the app offline.	T-02 , T-14

8. Edge cases and failure semantics

ID	Situation	Required behavior
E-01	No model selected / empty prompt	Run disabled; inline message; no worker spun.
E-02	A model fails mid-stream (raises inside the iterator, or times out)	Panel -> ERROR/TIMED_OUT ; partial text kept; siblings continue ; the run still settles. Never fabricate completion.
E-03	Structured parse or validation fails	Retry up to <code>max_retries</code> with an error-informed prompt; on exhaustion -> ERROR showing the first failure reason. Never present an unvalidated dict as VALID (I-009).
E-04	Streaming disabled (non-streaming path)	<code>ttft_ms == total_latency_ms</code> ; <code>tps</code> computed over the single interval (guards as in I-005).
E-05	Zero completion tokens (empty response)	<code>tps == 0.0</code> (I-005); not <code>inf/nan</code> ; status may still be COMPLETED .
E-06	<code>max_tokens</code> = 0 or other invalid param	Validation (§5.2) blocks Run; message shown.
E-07	Uncaught worker exception	Emit <code>crashed(model_id, msg)</code> ; panel -> ERROR ; never abort the whole process.

ID	Situation	Required behavior
E-08	Cancel while active	Every worker torn down and joined; no live worker survives (I-010); clean exit.
E-09	Unknown model id requested	<code>ModelRegistry.get</code> raises <code>KeyError</code> -> surfaced as panel <code>ERROR</code> , not a crash.
E-10	No display (CI)	Tests use <code>QT_QPA_PLATFORM=offscreen</code> ; logic unaffected.
E-11	Code-fence-wrapped JSON	<code>parse_json</code> strips a single json ... fence before loads; bare JSON also accepted.
E-12	Duplicate concurrent runs from the UI	Cancel any prior live run first; exactly one active run at a time (I-010).
E-13	Ollama daemon unreachable (not running / <code>OLLAMA_HOST</code> wrong)	<code>OllamaClient.list_models()</code> raises; on launch the UI falls back to the MockModel registry with a status banner <code>Ollama unavailable - using mock models</code> (R-16); no crash, no hang.
E-14	Model not pulled locally (Ollama returns <code>HTTP 404/model not found</code>)	That panel -> <code>ERROR</code> with <code>model not found: 'foo'</code> - pull it with <code>ollama pull</code> . Siblings continue; no crash (R-17 / E-09).
E-15	Malformed NDJSON line / missing <code>eval_count</code> in an Ollama stream	Skip non-JSON keepalive lines; on a malformed final line, keep partial text and set <code>usage</code> to best-effort <code>count_tokens</code> on the accumulated text (or 0) rather than crash; surface a warning on the panel (E-02).

Failure philosophy (spec-engineering doctrine): the model is the *probabilistic* side of the reliability boundary (§15); everything in §4 [C-04](#)/[C-05](#) is the *deterministic* side. The dominant failure here is **accepting an unvalidated artifact as valid** and **letting one model's failure poison the others**. This spec forbids both: [I-009](#) enforces the validate gate; [E-02](#)/[E-07](#) isolate faults per panel.

9. Acceptance criteria, tests, and evals

All tests target Level-3 executable criteria. Pure layers ([C-04](#), [C-05](#)) and mock models need **no Qt and no network**; GUI tests run offscreen via `pytest-qt`.

9.1 Model interface + mock model (deterministic, no Qt/network)

ID	Criterion
T-01	Usage arithmetic (I-001): <code>Usage(p,c).total_tokens == p + c</code> for representative pairs.
T-02	Interface purity (I-002 / K-04): a source scan asserts the <i>only</i> provider-aware module is <code>OllamaClient</code> ; no module imports a vendor SDK. <code>httpx</code> is allowed only inside <code>OllamaClient</code> . The app imports + runs with no Ollama and no keys (via <code>MockModel</code>).
T-07	Reproducibility (I-012 / I-008 / R-15): <code>MockModel</code> with <code>seed=42</code> yields bitwise-identical <code>text</code> and <code>usage</code> across two runs; its streamed token count equals its non-streaming <code>usage.completion_tokens</code> . Determinism of TTFB/latency is not asserted (wall-clock), but token counts are.

9.2 Metrics (pure) — fast, run always

ID	Criterion
T-03	Pricing location (I-003): changing a model's registry price changes <code>cost_usd</code> in the UI/worker with no other edit.
T-04	TTFT ordering & TPS guard (I-004, I-005): <code>ttft_ms</code> $\leq$ <code>total_latency_ms</code> ; zero-generation-duration $\Rightarrow$ <code>tps</code> $==$ 0.0 (not <code>inf/nan</code>).
T-05	Cost formula (I-006): <code>cost_usd</code> matches $N_{in}P_{in} + N_{out}P_{out}$ exactly for non-trivial prices.
T-06	Cost-per-task (I-007): all-failed run divides by 1; mixed run divides by the success count.

9.3 Structured output (pure) — the reliability boundary

ID	Criterion
T-08	Validate gate (I-009 / R-10): (a) a conforming object validates ok; (b) an out-of-range confidence (e.g. 1.5) fails ; (c) a missing required field fails ; (d) extra property fails (<code>additionalProperties:false</code>); (e) after <code>max_retries</code> exhaustion the panel is ERROR , never VALID .
T-14	Fence + bare JSON (E-11): <code>parse_json</code> accepts both <code>''\n{...}\n''</code> and bare <code>{...}</code> ; both route to the same schema; a non-JSON string returns (<code>None</code> , <code>reason</code>).

9.4 GUI / integration (offscreen, pytest-qt)

ID	Criterion
T-09	Single worker / cancel (I-010 / E-12): starting a run while one is active cancels the prior; after <code>cancel()</code> zero workers are alive.
T-10	Isolated failure (E-02 / K-02): with two mock models, one forced to raise mid-stream, that panel is ERROR while the sibling is COMPLETED ; the run settles.
T-11	Off-thread + liveness (I-011 / K-01): while a run streams offscreen, a posted task is serviced < 50 ms; the worker thread id $\neq$ the main thread id.
T-13	State machine (§3.1/§3.2): idle $\rightarrow$ run $\rightarrow$ (all settle) $\rightarrow$ idle; cancel from running reaches idle with terminal panels; control-enable flags match §3.1.
T-15	Validation UI (E-01): empty prompt and zero models selected disable Run with the inline messages.

9.5 Manual / smoke eval (not automated; recorded)

- Launch with `uv run model-playground`; confirm 2–3 models side by side: one **local Ollama** model (e.g. `mock/fast`, or a real `/api/tags` model if Ollama is running), one slow, one misbehaving (raises), and observe independent panels, streaming text, metric columns, and a structured pass/fail badge (E-02, qualitative).
- Confirm the Ollama-unreachable banner: with no daemon, the checklist falls back to the `MockModel` registry (E-13/R-16).

9.6 Ollama client (integration, no Qt; network-stubbed)

ID	Criterion
T-16	Discovery + fallback (R-16 / E-13): with <code>OllamaClient</code> pointed at an unreachable host, <code>list_models()</code> raises a connection error and the app falls back to <code>MockModel</code> — asserted by a <code>conftest</code> fixture that forces <code>OLLAMA_HOST</code> to a dead port; no network is real.
T-17	NDJSON + param mapping (C-03b / I-008 / E-15): using <code>httpx.MockTransport</code> (no real network), <code>OllamaClient.chat</code> (a) maps <code>max_tokens</code> → <code>num_predict</code> , <code>seed</code> → <code>seed</code> , <code>temperature/top_p</code> through; (b) parses a multi-line NDJSON stream into <code>StreamChunks</code> whose final chunk carries <code>finished=True</code> and <code>usage</code> from <code>eval_count/prompt_eval_count</code> ; (c) keeps partial text and best-effort usage when a trailing line is malformed.
T-18	Unknown model (R-17 / E-14): a mock 404/model not found response yields panel <code>ERROR</code> with the <code>model not found: '...'</code> message; it does not crash.

10. Dependencies and environment

Concern	Decision	Rationale
Package/env manager	uv	Fast, reproducible, pins Python; satisfies R-14.
Python	3.12 ($\geq 3.12, < 3.13$)	Requested; stable PyQt5.
GUI	PyQt5 (5.15)	Requested
HTTP	httpx (stream-capable)	Thin client for <code>OllamaModel</code> (C-03b) over the Ollama localhost API; not imported by pure layers (metrics/structured) nor by <code>MockModel</code> ; not required to run tests (K-04).
Local inference engine	Ollama (<i>external</i>)	Runtime at <code>http://localhost:11434</code> (§0 deployment decision). Not a Python dependency — a host prerequisite. The app degrades to <code>MockModel</code> when it is absent (E-13).
Schema validation	jsonschema	C-05 validate.
Mock model	in-repo MockModel	Deterministic, offline, reproducible (R-15); the default test/runtime double; provides streaming + a “slow” and a “raising” variant for E-02/E-07.
Dev deps	pytest, pytest-qt, ruff	Automated Level-3 tests (§9) + lint.
GUI test backend	QT_QPA_PLATFORM=offscreen	Headless CI (E-10).

Reproducibility (see `README.md`):

```
# host prerequisite for the real path (optional; the mock path needs neither):
#  ollama pull llama3.2
uv sync                                # create .venv (Python 3.12), install everything
uv run pytest                          # run the §9 suite (fully offline; no Ollama needed)
uv run model-playground                # launch GUI (Ollama if reachable, else mock models)
```

11. Traceability matrix (id -> where realized)

R-01	--> C-02(Model), C-03(Registry)	--> T-02
R-02	--> C-01(GenerationParams), C-08 controls	--> T-15
R-03	--> C-06(token signal), C-07 panel	--> T-11
R-04	--> C-04(RunMetrics)	--> T-04
R-05	--> C-01(Usage), I-001	--> T-01
R-06	--> C-03(cost_usd), I-003/006	--> T-03, T-05
R-07	--> §5.1 grid, C-06 N workers	--> T-10, T-13
R-08	--> C-04(ttft/tps), §3.4	--> T-04
R-09/10	--> C-05(parse,validate), I-009	--> T-08
R-11	--> §8 E-03, C-05 retry	--> T-08
R-14	--> §10, pyproject, K-04	--> T-02
R-15	--> C-01(seed), MockModel	--> T-07
I-007 / §13	--> C-04 cost-per-task	--> T-06
I-010 / E-12	--> C-06(cancel), §3.2	--> T-09
I-011 / K-01	--> §3.3 worker threads	--> T-11
E-02 / K-02	--> §3.2 per-model terminal	--> T-10
R-05 / R-15	--> C-01(Usage: Ollama eval_count / prompt_eval_count), I-001/008/012	--> T-01, T-07
R-16 / E-13	--> C-03b(list_models), C-03 discovery, §8	--> (smoke, E-13)
R-17 / E-14	--> C-03b(chat 404 -> ERROR)	--> T-10-style
C-03b / I-002	--> OllamaClient (only provider-aware module)	--> T-02
C-03 / I-003	--> local default price 0.0	--> T-05
E-15	--> C-03b(NDJSON parse: keep partial + best-effort usage)	--> (smoke)

Open questions / ambiguities flagged for the human (spec elicitation):

1. **Structured schema.** The default schema is the §15 “answer” object. Should it be *user-editable* in the UI, or fixed to this one object for v0.1? (spec: fixed for v0.1; Q-01).
2. **Pricing source.** Prices are hand-entered per model in the registry for v0.1. Should we pull a canonical price table? (Q-02 — recommend: hand-entered, documented as illustrative.)
3. **Default local model.** Which Ollama model(s) should the GUI pre-select for the first smoke run? (Q-03 - recommend: pre-select the first model returned by `/api/tags`, plus the built-in `mock/fast` and `mock/slow`.)
4. **Tool calling.** §7 (tool calling) is deliberately **out of scope** here, as is §8 (multimodal). Confirm these remain conceptual-only in v0.1. (Q-04)
5. **Concurrency default.** Concurrent (one worker/model) vs. sequential default. Spec picks **concurrent**, with a sequential checkbox. (Q-05 — confirm.)
6. **Retry semantics for non-structured mode.** Retries apply only to the structured pipeline; plain-text mode has no retry. Confirm. (Q-06)

End of specification. This document is the source of truth; implementation and tests are to be derived from it and kept in sync per §11.